NON-MEASUREMENT SENSITIVE

MIL-DTL-46725D(AT)
20 August 1998
SUPERSEDING
MIL-PRF-46725C(AT)
1 November 1995

DETAIL SPECIFICATION

VALVE, FOUR-WAY, HYDRAULIC, MANUAL

This specification is approved for use by the U.S. Army Tank-automotive and Armaments Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers two types of manually operated four-way hydraulic directional control valves of 20 684 kilopascals (kPa) [3000 pounds per square inch (psi)] operating pressure (see 6.1).

1.2 Classification. Valves will be of the following types, and classifications as specified (see 6.2):

Type I	-	Four-Way, Two-Position, 7.6 liters per minute (L/min) [2 gallons per minute (gal/min)] flow capacity.
Type II	-	Four-Way, Three-Position.
Class A	-	7.6 L/min (2 gal/min) flow rate.
Class B	-	22.7 L/min (6 gal/min) flow rate.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Tank-automotive and Armaments Command, ATTN: AMSTA-TR-E/BLUE, Warren, MI 48397-5000 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 2590

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T06:47Z
Check the source to verify that this is the current version before use.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for addition information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirement documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplements thereto cited in the solicitation (see 6.2).

SPECIFICATIONS

DEPARTMENT OF DEFENSE

MIL-PRF-6083 - Hydraulic Fluid, Petroleum Base: For Preservation and Operation.

STANDARDS

DEPARTMENT OF DEFENSE

MIL-STD-810 - Environmental Test Methods and Engineering Guidelines (see 4.3.1).

(Unless otherwise indicated, copies of the above specifications, standards, and handbooks are available from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.2.2 Other Government documents, drawings, and publications. The following other Government document, drawing, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T06:47Z
Check the source to verify that this is the current version before use.

DRAWINGS

ARMY

9093264	- Valve, Two-Position, Four-Way.
9094352	- Valve, Three-Position, Four-Way.
9098279	- Valve, Three-Position, Four-way.

(Copies of these drawings are available from the U.S. Army Tank-automotive and Armaments Command, ATTN: AMSTA-TR-E/BLUE, Warren, MI 48397-5000.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DoDISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS are the issue of the documents cited in the solicitation (see 6.2).

GENERAL MOTORS CORPORATION

GM 9540P	- Accelerated Corrosion Test.

(Copies of GM publications may be obtained from General Motors Corporation, C/O Global Engineering, 15 Inverness Way, Englewood, CO 80112.)

2.4 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.1.1.

3.2 Design, materials, and manufacturing processes. Unless otherwise specified herein, the design, materials, and manufacturing process selection shall be the prerogative of the contractor as long as all articles submitted to the government fully meet the operating, interface, support and ownership, and operating environment requirements specified.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T06:47Z
Check the source to verify that this is the current version before use.

3.2.1 Construction. The valve shall be constructed to the form and dimensions shown on the applicable drawing listed in table I. The shape is optional within the maximum dimensions.

TABLE I. Applicable drawings.

Valve Type	Valve Class	Drawing Number
I	-	9093264
II	A	9098279
	B	9094352

3.2.2 Recycled, recovered, or environmentally preferable materials. Recycled, recovered, or environmentally preferable materials should be used to the maximum extent possible provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.3 Operating requirements. Each valve shall provide the following functional, operational, and performance capabilities.

3.3.1 Flow rate and pressure drop.

3.3.1.1 Type I. The flow rate shall be 7.6 L/min (2 gal/min) through any interconnected ports with no more than 1034 kPa (150 psi) pressure drop across the valve.

3.3.1.2 Type II. The flow rate of class A valves shall be 7.6 L/min (2 gal/min) through any interconnected ports, with less than 689 kPa (100 psi) pressure drop across the valve. The flow rate of class B valves shall be 22.7 L/min (6 gal/min) with less than 1724 kPa (250 psi) pressure drop across the valve.

3.3.2 Operating torque.

3.3.2.1 Type I. The maximum operating torque shall be 5.6 Newton-meters (N-m) [50 pound-inches (lb-in.)] with 20 684 kPa (3000 psi) applied at the pressure port.

3.3.2.2 Type II. The maximum operating torque shall be 2.8 N-m (25 lb-in.) for class A valves, and 5.9 N-m (52 lb-in.) for class B valves, with 20 684 kPa (3000 psi) hydraulic pressure applied at the pressure port.

3.3.3 Normal operating pressure. The valve shall operate satisfactorily throughout a hydraulic pressure range of 0 to 20 684 kPa (3000 psi) applied to any port, except the return port, while using hydraulic fluid conforming to MIL-PRF-6083.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T06:47Z
Check the source to verify that this is the current version before use.

3.3.4 Proof pressure. The valve shall be capable of operating for up to 2 minutes at a hydraulic proof pressure of 31 026 kPa (4500 psi), except for the return port which shall be capable of operating at 3447 kPa (500 psi) for type I valves and 10 342 kPa (1500 psi) for type II valves without external leakage, permanent deformation, or failure.

3.3.5 Burst pressure. The valve shall withstand a hydraulic pressure of 51 711 kPa (7500 psi), except for the return ports. Without bursting, return ports shall withstand 5723 kPa (830 psi) for type I valves and 17 237 kPa (2500 psi) for type II valves.

3.3.6 Valve spool. The valve spool shall be hydraulically balanced so that the forces required to manipulate the levers shall not vary more than 10 percent (%) due to variation in pressure and flow. Spools shall not bind under any operating conditions specified herein.

3.3.7 Leakage. The valve shall be capable of operating at 20 684 kPa (3000 psi) hydraulic pressure applied to sealing members of the individual fluid chambers without external leakage, and with a combined internal leakage from the ports not exceeding 10 cubic centimeters per minute (cm^3/min).

3.3.8 Life. The valve shall be capable of operating for 20 000 cycles at 20 684 kPa (3000 psi) hydraulic working pressure, without external leakage.

3.4 Interface requirements.

3.4.1 Interchangeability. All parts having the same manufacturer's part number shall be directly and completely interchangeable with each other with respect to installation and performance. Subassemblies, composed of mating components, shall be interchangeable as assembled units.

3.4.2 Weight. The gross weight of the valve shall not exceed 0.9 kilograms (kg) (2 pounds (lb)) for type I valves and 1.1 kg (2.5 lb) for type II valves.

3.5 Support and ownership requirements.

3.5.1 Finish. Unless otherwise specified (see 6.2), interior surfaces of all parts of each valve shall be flushed with rust-inhibiting fluid.

3.5.2 Identification marking. Markings shall be permanent and legible. When specified in the contract or order (see 6.2), nameplates shall be included. Nameplates shall be made of material suitable to withstand the environmental requirements specified herein and remain legible. Marking shall include the following information in the order shown:

a. Valve, 4-way, hydraulic, manual
b. National Stock Number (NSN)
c. Part or identifying number (PIN)

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T06:47Z
Check the source to verify that this is the current version before use.

d. Manufacturer's Serial Number
e. Manufacturer's name, identification code (CAGE) or trademark
f. Contract or order number
g. Date of manufacture
h. US

3.5.3 Safety. The valve shall be free of foreign materials, and shall pose no hazards physically to operators.

3.6 Operating environmental requirements. The valve shall operate under the following environmental conditions, without degradation in performance.

3.6.1 High temperature. The valve shall perform in accordance with requirements of 3.3.1 and 3.3.8 while operating at 71 ± 3 degrees Celsius (°C) [160 ± 5 degrees Fahrenheit (°F)].

3.6.2 Low temperature. The valve shall perform in accordance with requirements of 3.3.1 and 3.3.8 while operating at -54 ± 3°C (-65 ± 5°F).

3.6.3 Fungus. The valve shall operate in environments conducive to spore germination and growth (air temperature between 30° ± 1°C (86° ± 2°F) and relative humidity of 95 ± 5%).

3.6.4 Corrosion. The valve and the removed valve spool shall be capable of withstanding a 1 % salt spray solution.

3.6.5 Dust. The valve shall be capable of withstanding a blowing dust environment with an air velocity of 1.5 to 8.9 meter per second (m/s) (300 to 1750 feet per minute (ft/min)).

3.6.6 Shock. The valve shall operate after a half sine shock pulse of 15 gravity units (g).

3.6.7 Vibration. The valve components shall not loosen nor become damaged after being subjected to a simple harmonic motion having an amplitude of 0.8 millimeter (mm).

4. VERIFICATION

4.1 Classification of inspection. The inspection conditions specified herein are classified as follows:

a. First article inspection (see 4.1.1).
b. Conformance inspection (CI) (see 4.1.2).

4.1.1 First article inspection. When specified, first article inspection shall be performed on 4 preproduction or initial production valves (see 6.2 and 6.4). This inspection shall include all verifications listed in table II.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T06:47Z
Check the source to verify that this is the current version before use.

4.1.2 Conformance inspection. Conformance inspection shall include the examinations and tests listed in table II.

4.1.2.1 Sampling. Samples for examinations and tests shall be specified in the contract or order (see 6.2).

4.1.2.2 Examinations. Samples selected shall be examined in accordance with table II.

4.1.2.3 Tests. Samples selected shall be tested in accordance with table II.

4.2 Order of inspection. The inspection sequence may be in any order, except burst pressure shall be last.

4.3 Verification methods. The types of verification methods included in this section are visual inspection, measurement, sample tests, full-scale demonstration tests, simulation, modeling, engineering evaluation, component properties analysis, and similarity to previously-approved or previously-qualified designs.

4.3.1 Verification alternatives. The manufacturer may propose alternative test methods, techniques, or equipment, including the application of statistical process control, tool control, or cost-effective sampling procedures, to verify performance. See the contract for alternatives that replace verifications required by the specifications.

4.3.2 Inspection conditions. Unless otherwise specified (see 6.2), all inspections shall be conducted under the following standard (room) ambient conditions:

a. Ambient air temperature: 23 ± 10°C (77 ± 15°F)
b. Barometric pressure: 725 (+50, -75) millimeters (mm) mercury (Hg) [28.5 (+2, -3) inches (in.)] Hg)
c. Relative humidity: 50 ± 30 percent.

Unless otherwise specified herein, the valve shall meet all specified performance requirements in a hydraulic system, utilizing hydraulic fluid conforming to MIL-PRF-6083 under the following conditions:

a. Hydraulic fluid temperature of 38 ± 5°C (70 ± 8°F).
b. All grease or foreign matter shall be removed from the interior parts of the valve before testing.
c. Air shall be completely removed from all hydraulic portions of the valve prior to test.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T06:47Z
Check the source to verify that this is the current version before use.

TABLE II. Verification methods.

Title	Requirement	Verification	First article	CI	
				Examin-ation	Test
Design, materials and manufacturing processes	3.2	4.3.3	X		
Operating requirements	3.3	4.3.4			
Flow rate and pressure drop	3.3.1	4.3.4.1			
Type I	3.3.1.1	4.3.4.1	X		X
Type II	3.3.1.2	4.3.4.1	X		X
Operating torque	3.3.2	4.3.4.2			
Type I	3.3.2.1	4.3.4.2	X		X
Type II	3.3.2.2	4.3.4.2	X		X
Normal operating pressure	3.3.3	4.3.4.3	X		X
Proof pressure	3.3.4	4.3.4.4	X		
Type I	3.3.4	4.3.4.4			
Type II	3.3.4	4.3.4.4	X		
Burst pressure	3.3.5	4.3.4.5	X		
Valve spool	3.3.6	4.3.3			
Leakage	3.3.7	4.3.4.6	X		X
Life	3.3.8	4.3.4.7	X		
Interface requirements	3.4	4.3.5			
Interchangeability	3.4.1	4.3.5	X	X	
Weight	3.4.2	4.3.5.1	X	X	
Support and ownership requirements	3.5	4.3.6			
Finish	3.5.1	4.3.6.1			
Identification marking	3.5.2	4.3.6.2			
Safety	3.5.3	4.3.6.3			
Operating environmental requirements	3.6	4.3.7			
High temperature	3.6.1	4.3.7.1	X		
Low temperature	3.6.2	4.3.7.2	X		
Fungus	3.6.3	4.3.7.3	X		
Corrosion	3.6.4	4.3.7.4	X		
Dust	3.6.5	4.3.7.5	X		
Shock	3.6.6	4.3.7.6	X		
Vibration	3.6.7	4.3.7.7	X		

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T06:47Z
Check the source to verify that this is the current version before use.

4.3.3 Material, design and construction. Use one or more of the methods outlined in 4.3 or 4.3.1 to verify that the valve is built in accordance with Drawings 9093264, 9098279, and 9094352. Conformance shall be determined by inspection of the contractor records providing proof or certification that design, construction, processing, and materials conform to requirements. Applicable records shall include drawings, specifications, design data, receiving inspection records, processing and quality control standards, vendor catalogs and certifications, industry standards, test reports, and rating data.

4.3.4 Operating requirements verification.

4.3.4.1 Flow rate and pressure drop test. Verify that the flow rate of the valve shall be accurately maintained at 7.6 L/min (2 gal/min) with no more than 1034 kPa (150 psi) pressure drop across the valve for type I, 7.6 L/min (2 gal/min) with less than 689 kPa (100 psi) for class A type II, and 22.7 L/min (6 gal/min) with less than 1724 kPa (250 psi) for class B type II, using a manometer to measure pressure drop from one port to another. Failure to maintain the specified flow rate shall constitute failure of the test.

4.3.4.2 Operating torque test. With a hydraulic pressure of 20 684 kPa (3000 psi) at the pressure port, the torque required to operate the valves shall be 5.6 N-m (50 lb-in.) for type I, 2.8 N-m (25 lb-in.) for class A type II, and 5.9 N-m (52 lb-in.) for class B type II. Measurement outside the specified range shall constitute failure of the test.

4.3.4.3 Operating pressure test. Pressure shall be applied to each port, except the return port, and varied throughout a range of 0 psi to 20 684 kPa (3000 psi). Operation shall be observed. Failure of the valve to operate satisfactorily throughout the specified range shall constitute failure of the test.

4.3.4.4 Proof pressure. Proof pressure of 31 026 kPa (4500 psi) shall be applied at the pressure and return ports for a period of 2 minutes. The valve shall be manually actuated after application of pressures. There shall be no evidence of external leakage, permanent deformation or failure.

4.3.4.5 Burst pressure. This test shall be performed after completion of all other tests. The valve shall be subjected to a hydraulic pressure of 51 711 kPa (7500 psi) for the pressure ports for a period of 3 minutes. The valve shall be subjected to 5723 kPa (830 psi) for type I valves and 17 237 (2500 psi) for type II valves for the return port, for a period of 3 minutes at each application without bursting. This test shall be performed for all three positions. When testing pressure ports, the return port shall be open. The rate of application of pressure shall not exceed 179 267 kPa (26 000 psi) per minute. The valve need not be operable after these tests.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T06:47Z
Check the source to verify that this is the current version before use.

4.3.4.6 Leakage. The valve shall be tested for leakage by applying a hydraulic pressure of 20 684 kPa (3000 psi) to each sealing member of individual fluid chambers, in succession for a 2 minute period each. The pressure shall then be lowered to 34 to 69 kPa (5 to 10 psig) and held for 2 minutes. There shall be no external leakage.

4.3.4.7 Life test. The valve shall be subjected to a hydraulic pressure of 20 684 kPa (3000 psi) with a flow rate of 7.6L/min (2 gal/min) for both type I and type II (class A), and 22.7 L/min (6 gal/min) for type II (class B), for 20 000 cycles. Each cycle shall consist of operating the valve from one extreme position to the other extreme position and return to the starting position. The valve shall be observed for evidence of external leakage during cycling. After cycling, internal leakage shall not exceed 10 cm^3/min. The valve shall evidence no excessive wear or breakdown.

4.3.5 Interface requirements verification. Use one or more of the methods outlined in 4.3 or 4.3.1 to verify the interchangeability of like parts.

4.3.5.1 Weight inspection. Weigh the valve to be less than 0.9 kg (2 lb) for type I and 1.1 kg (2.5 lb) for type II

4.3.6 Support and ownership requirements verification.

4.3.6.1 Finish. Flush valve with rust inhibiting fluid. Verify that all parts are covered.

4.3.6.2 Identification marking. Verify the presence of the required marking. After all environmental testing, re-inspect marking for legibility.

4.3.6.3 Safety. Visually inspect valve for burrs, chips, sharp edges, or other imperfections that pose physical danger to an installer/operator.

4.3.7 Environmental requirements verification.

4.3.7.1 High temperature. The valve shall be placed within a test chamber and exposed to a temperature of 71°C (160°F) for 24 hours duration. After 24 hours of exposure, the valve shall be charged to 20 684 kPa (3000 psi) hydraulic pressure and flow rate (see 3.3.1) which shall be maintained for a period of 4 hours. At the conclusion of the exposure period, the valve shall be tested as specified in 4.3.4.7 for 50 cycles. After testing, cool the valve to room temperature and verify that it passes the tests of 4.3.4.1, 4.3.4.6, and 4.3.4.7 for 50 cycles.

4.3.7.2 Low temperature . The valve shall be placed within a test chamber and exposed to a temperature of -54°C (-65°F) for 24 hours duration. After 24 hours of low temperature exposure, the valve shall be charged to a hydraulic pressure of 20 684 kPa (3000 psi), and a flow

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T06:47Z
Check the source to verify that this is the current version before use.

rate as specified (see 3.3.1) shall be maintained for a period of 4 hours. At the conclusion of the 4-hour period, the valve shall be tested as specified in 4.3.4.7 for 50 cycles. After testing stabilize the valve at room temperature and verify that it passes the tests of 4.3.4.1, 4.3.4.6, and 4.3.4.7 for 50 cycles.

4.3.7.3 Fungus. The valve shall be subjected to the fungus test specified in method 508.4 of MIL-STD-810 or equivalent. Upon completion of the test the valve shall be subjected to and pass the tests of 4.3.4.1, 4.3.4.6, and 4.3.4.7 for 50 cycles.

4.3.7.4 Corrosion. The valve assembly shall be subjected to 40 cycles of Accelerated Corrosion Test GM 9540P, method B. Following the exposure the valve shall be subjected to and pass the tests of 4.3.4.1, 4.3.4.6, and 4.3.4.7 for 50 cycles.

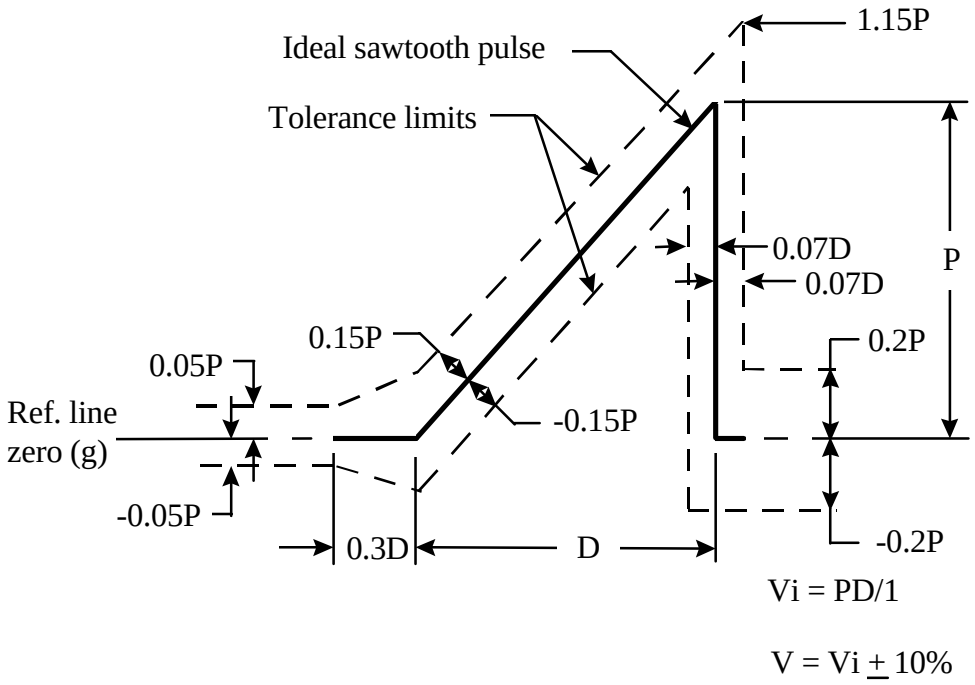
4.3.7.4.1 Valve spool corrosion. The valve spool shall be removed from the assembly, tested as specified in 4.3.4.1, 4.3.4.6, and 4.3.4.7 and subjected to 20 cycles of Accelerated Corrosion Test GM 9540P, method B. After testing, the valve spool shall exhibit no corrosion in order to meet the corrosion requirement of 3.6.4.

4.3.7.5 Dust. The valve shall be subjected to the blowing dust test (procedure I) specified in method 510.3 of MIL-STD-810 or equivalent, except that the maximum temperature shall be 52°C (+125°F). Following the test, the valve shall be subjected to and pass the tests of 4.3.4.1, 4.3.4.6, and 4.3.4.7 for 50 cycles.

4.3.7.6 Shock. The valve shall be subjected to a half sine shock pulse of 15 gravity units (g) for a period of 11 milliseconds (see figure 1). Following the test, the valve shall be examined for evidence of damage and shall be subjected to and pass the tests of 4.3.4.1, 4.3.4.6, and 4.3.4.7 for 50 cycles.

4.3.7.7 Vibration. The valve shall be subjected to a simple harmonic motion having an amplitude of 0.8 mm (0.03 in.) [1.5 mm (0.06 in.) maximum total excursion), the frequency being varied uniformly between the approximate limits of 10 and 55 hertz (Hz). The entire frequency range, from 10 to 55 Hz and return to 10 Hz, shall be traversed in approximately 1 minute. The vibration shall be for 1 hour at 25 $\pm$ 8°C (77 $\pm$ 15°F) and 30 minutes at -54°C (-65 $\pm$ 5°F) in each direction of the three major axes. Following the vibration, the valve shall be examined for evidence of damage and shall be subjected to and pass the tests of 4.3.4.1, 4.3.4.6, and 4.3.4.7 for 50 cycles.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T06:47Z
Check the source to verify that this is the current version before use.



NOTE:

The oscillogram should include a time about 3D long with the pulse located approximately in the center. The integration to determine velocity change should extend from 0.4D before the pulse to 0.1D beyond the pulse. The acceleration amplitude of the ideal half sine pulse is A and its duration is D. Any measured acceleration pulse which can be contained between the broken line boundaries is a nominal half sine pulse of nominal amplitude A and nominal duration D. The velocity-change associated with the measured acceleration pulse is Vi.

FIGURE 1. Tolerances for half sine shock pulse.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of materiel is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department of Defense Agency, or within the Military Department's System Command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T06:47Z
Check the source to verify that this is the current version before use.

6. NOTES

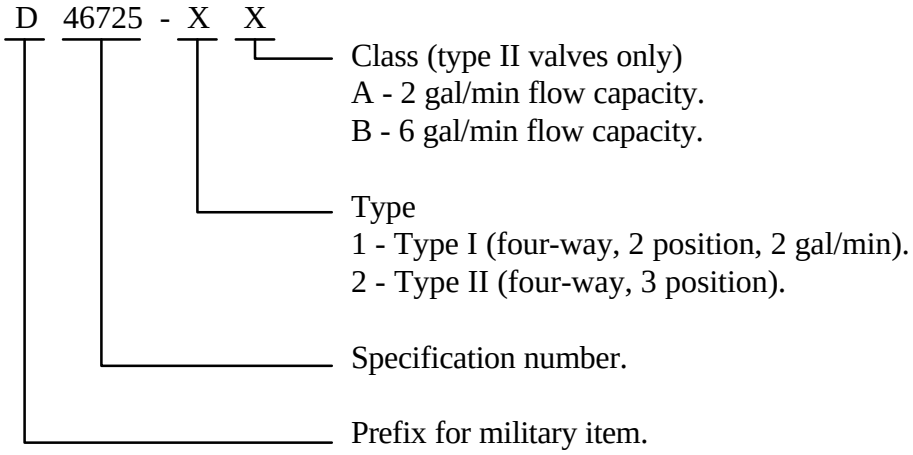
(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The valves covered by this performance specification are intended to be used to manually select the fluid flow paths in a 3000 psi hydraulic system in order to control cylinder actuation. They are intended for use with the loader-transporter, guided missile (Hawk-Loader). These valves are military unique because they must operate satisfactorily throughout a hydraulic range of 0 to 20 684 kPa (3000 psi), and at a temperature range from - 65°F to + 160°F which exceed any commercial ranges.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.
b. Type and class of valve required (see 1.2).
c. Issue of DoDISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.2.1 and 2.3).
d. When first article is required and if samples are to be preproduction or initial production(see 3.1 and 4.1.1).
e. Valve finish if other than as specified (see 3.5.1).
f. If nameplate is to be included (see 3.5.2).
g. Sampling requirements (see 4.1.2.1).
h. Test conditions, if other than as specified (see 4.3.2).
i. Packaging requirements (see 5.1).
j. PIN and quantity required (see 6.3).

6.3 Part or identifying number (PIN). The pins to be used for valves acquired to this specification are established as follows:



Source: https://assist.dla.mil -- Downloaded: 2026-10-22T06:47Z
Check the source to verify that this is the current version before use.

6.4 First article. When requiring a first article inspection, contracting documents should provide specific guidance to offerors. This guidance should cover whether the first article is a first article sample, a first production item, or the number of test items. These documents should also include specific instructions regarding arrangements for examinations, approval of first article test results, and disposition of first articles. Pre-solicitations documents should provide Government waiver rights for samples for first article inspection to bidders offering a previously acquired or tested product. Bidders offering such products who wish to rely on such production testing must furnish evidence with the bid that prior Government approval is appropriate for the pending contract.

6.5 Conformance inspection. Affordable conformance inspection with confidence varies depending upon a number of procurement risk factors. Some of these factors include: Contractor past performance, government schedules and budget, product material and design maturity, manufacturing capital equipment and processes applied, the controlled uniformity of those processes, labor skill and training, and the uniformity of measuring processes and techniques. During the solicitations, contracting documents should indicate those tests desired from table II and their designated frequency based on a risk assessment for the procurement.

6.6 Subject term (key word) listing.

Flow	Spool
Hawk-Loader	Torque
Pressure	

6.7 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
Army - AT

Preparing Activity:
Army - AT

(Project 2590-0261)

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T06:47Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	**1. DOCUMENT NUMBER** MIL-DTL-46725D(AT)	**2. DOCUMENT DATE *(YYMMDD)*** 980820

3. DOCUMENT TITLE

Valve, Four-Way, Hydraulic, Manual

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME *(Last, First, Middle Initial)*	**b. ORGANIZATION**	
c. ADDRESS *(Include Zip Code)*	**d. TELEPHONE *(Include Area Code)*** **(1) Commercial** **(2) AUTOVON** ***(If applicable)***	**7. DATE SUBMITTED *(YYMMDD)***

8. PREPARING ACTIVITY

a. NAME	**b. TELEPHONE *(Include Area Code)*** **(1) Commercial** **(2) AUTOVON** **(810) 574-8745** **786-8745**
c. ADDRESS *(Include Zip Code)* **Commander** **U.S. Army Tank-automotive and Armaments Command** **ATTN: AMSTA-TR-E/BLUE** **Warren, MI 48397-5000**	**IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:** **Defense Quality and Standardization Office** **5203 Leesburg Pike, Suite 1403** **Falls Church, VA 22041-3466** **Telephone (703) 756-2340 AUTOVON 289-2340**

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T06:47Z
Check the source to verify that this is the current version before use.